E2E Final Board

DB1

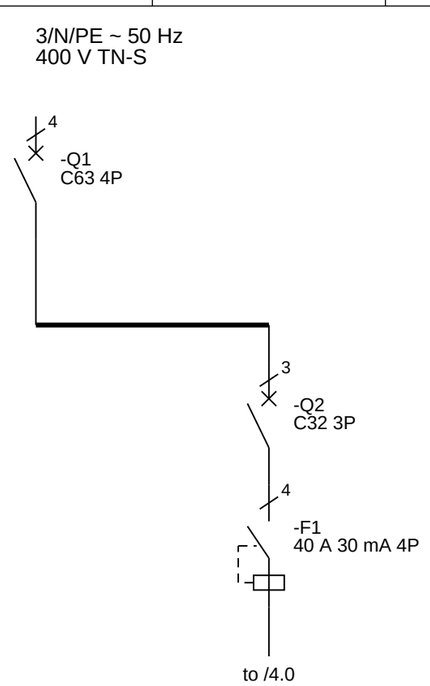
Customer	Test Customer
Company profile	iec-default
Design rules	software defaults, not confirmed by the company's engineers

Rev.	Date	Description	Drawn	Checked	Review approved
01	2026-10-23	First issue			
02	2026-10-23	AC raised to 3.5 kW			Eng. Browser Test

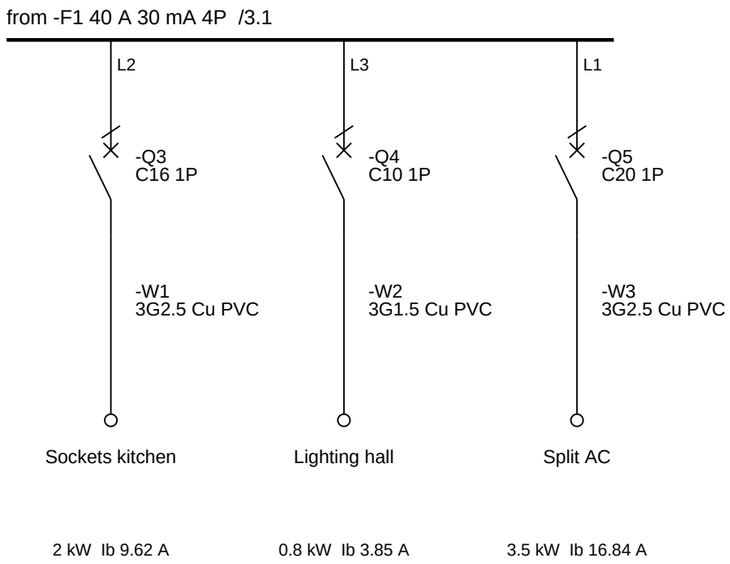
Company	Project E2E Final Board	Job number	Board	Title of page **Title page**	Rev. 02
Date 2026-10-23	Drawn	Checked	Review approved Eng. Browser Test	Page 1 of 9	

Page	Title	Board
1	Title page	
3	Main power	DB1
4	Distribution loads	DB1
5	Design notes	DB1
6	Layout	DB1
7	Terminals	DB1
8	Cable list	
9	Parts list	

Company	Project	Job number	Board	Title of page	Rev.
	E2E Final Board			**Table of contents**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-23			Eng. Browser Test	2 of 9	



Company	Project	Job number	Board	Title of page	Rev.
	E2E Final Board		DB1	**Main power**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-23			Eng. Browser Test	3 of 9	

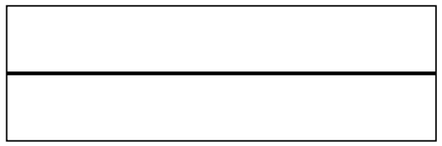


Company	Project	Job number	Board	Title of page	Rev.
	E2E Final Board		DB1	**Distribution loads**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-23			Eng. Browser Test	4 of 9	

No.	Note
1	Split AC: Ib 16.84 A exceeds the company's 16 A for air_conditioning; rated 20 A instead.
2	Group 1 breaker raised to 32 A, 1.6 x the 20 A breaker after it, for discrimination.
3	Phase imbalance 77.1 % exceeds the company's 10 %: the loads cannot be spread more evenly as given.
4	Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination.
5	No fault level given: every breaker's breaking capacity is left to be confirmed.
6	Leave 1 spare outgoing ways (20 %).
7	Loads without a power factor were taken at cos phi 0.9.
8	Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.
9	Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.
10	Designed for Palestine, under IEC 60364 (adopted by the Palestine Standards Institution): its stated limits apply beneath the company's own settings.
11	Circuit rules are this software's defaults, not confirmed by the company's engineers.
12	Enclosure: ABB Mini Center compact GDMS332 RX (2CVA250001P0001), 2 rows of 16 modules, H x W x D 498 x 400 x 117 mm, IP41, busbar 200/250 A, 35 kA; 18 modules are left for spare ways (ABB Distribution Boards catalogue (1SKC802027C0201), Mini Center compact, pp. 5, 7). The outgoing terminal strip is not counted.

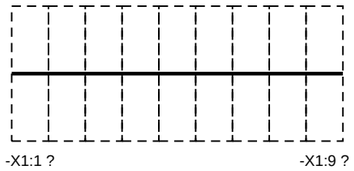
Company	Project	Job number	Board	Title of page	Rev.
	E2E Final Board		DB1	**Design notes**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-23			Eng. Browser Test	5 of 9	

Incomer: 70 mm of rail



Residual current group: 175 mm of rail

Terminals: 0 mm of rail; 9 without a width



Dashed: no width is given for the device's kind (rail_widths_mm in the company settings).

Company	Project	Job number	Board	Title of page	Rev.
	E2E Final Board		DB1	**Layout**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-23			Eng. Browser Test	6 of 9	

Terminal	Conductor	Circuit	Cable	mm²	Article
-X1:1	L2	Sockets kitchen	-W1	2.5	8WH1000-0AF00
-X1:2	N	Sockets kitchen	-W1	2.5	8WH1000-0AF00
-X1:3	PE	Sockets kitchen	-W1	2.5	8WH1000-0CF07
-X1:4	L3	Lighting hall	-W2	1.5	8WH1000-0AF00
-X1:5	N	Lighting hall	-W2	1.5	8WH1000-0AF00
-X1:6	PE	Lighting hall	-W2	1.5	8WH1000-0CF07
-X1:7	L1	Split AC	-W3	2.5	8WH1000-0AF00
-X1:8	N	Split AC	-W3	2.5	8WH1000-0AF00
-X1:9	PE	Split AC	-W3	2.5	8WH1000-0CF07

Company	Project	Job number	Board	Title of page	Rev.
	E2E Final Board		DB1	**Terminals**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-23			Eng. Browser Test	7 of 9	

Cable	From	To	Type	Length
DB1 -W1	DB1 -Q3	Sockets kitchen	3G2.5 Cu PVC	20 m
DB1 -W2	DB1 -Q4	Lighting hall	3G1.5 Cu PVC	20 m
DB1 -W3	DB1 -Q5	Split AC	3G2.5 Cu PVC	20 m

Company	Project	Job number	Board	Title of page	Rev.
	E2E Final Board			**Cable list**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-23			Eng. Browser Test	8 of 9	

Qty	Description	Article	Order number	Devices
1	Circuit-breaker C63 4P	not selected		-Q1
1	Circuit-breaker C32 3P	not selected		-Q2
1	Residual current circuit-breaker 40 A 30 mA 4P	not selected		-F1
1	Circuit-breaker C16 1P	not selected		-Q3
1	Circuit-breaker C10 1P	not selected		-Q4
1	Circuit-breaker C20 1P	not selected		-Q5
6	Terminal 8WH1	Siemens 8WH1000-0AF00	8WH1000-0AF00	DB1 -X1
3	PE terminal 8WH1	Siemens 8WH1000-0CF07	8WH1000-0CF07	DB1 -X1

Company	Project	Job number	Board	Title of page	Rev.
	E2E Final Board			**Parts list**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-23			Eng. Browser Test	9 of 9	

Findings (0 blocking, 5 warning, 7 info), page 1 of 1

[WARNING] DB1 / c3: Split AC: Ib 16.84 A exceeds the company's 16 A for air_conditioning; rated 20 A instead. (above_company_rating; source: policy; engineering confirmation required)
[WARNING] DB1: Phase imbalance 77.1 % exceeds the company's 10 %: the loads cannot be spread more evenly as given. (phase_imbalance; source: company max_phase_imbalance_percent; engineering confirmation required)
[WARNING] DB1: No fault level given: every breaker's breaking capacity is left to be confirmed. (no_fault_level; source: policy; engineering confirmation required)
[WARNING] DB1: Loads without a power factor were taken at cos phi 0.9. (default_power_factor; source: policy; engineering confirmation required)
[WARNING] DB1: Circuit rules are this software's defaults, not confirmed by the company's engineers. (rules_unconfirmed; source: policy; engineering confirmation required)
[INFO] DB1: Group 1 breaker raised to 32 A, 1.6 x the 20 A breaker after it, for discrimination. (discrimination_group_raised; source: policy)
[INFO] DB1: Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination. (discrimination_incomer_raised; source: policy)
[INFO] DB1: Leave 1 spare outgoing ways (20 %). (spare_ways; source: policy)
[INFO] DB1: Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables. (discrimination; source: policy)
[INFO] DB1: Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation. (cable_conditions; source: policy)
[INFO] DB1: Designed for Palestine, under IEC 60364 (adopted by the Palestine Standards Institution): its stated limits apply beneath the company's own settings. (market_rules; source: policy)
[INFO] DB1: Enclosure: ABB Mini Center compact GDMS332 RX (2CVA250001P0001), 2 rows of 16 modules, H x W x D 498 x 400 x 117 mm, IP41, busbar 200/250 A, 35 kA; 18 modules are left for spare ways (ABB Distribution Boards catalogue (1SKC802027C0201), Mini Center compact, pp. 5, 7). The outgoing terminal strip is not counted. (enclosure_selected; source: policy)